Reproducing VoxelMorph: Unsupervised Learning for Deformable Brain MRI Registration

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1 Introduction

MRI, as the most common technique in modern medicine, is used to assess patients' conditions before and after a series of treatments. Medical image registration is the task of spatially aligning two or more images of the same subject acquired at different times, from different scanners, or with different imaging parameters. The paper we are reproducing is VoxelMorph [1] (Balakrishnan et al., IEEE TMI 2019). We reproduce the core VoxelMorph framework using its official PyTorch implementation, train it on the OASIS-1 brain MRI dataset [4], and evaluate registration accuracy using the Dice Similarity Coefficient (DSC) on FSL brain segmentation labels [6]. We demonstrate consistent improvement over the unregistered baseline across all 20 test pairs, validating the paper's central claim.

Project Type: Reproducing Project.

CV Problem Types: Prediction (dense deformation field estimation), Recognition (anatomical correspondence across subjects), Detection (anatomical landmark alignment for evaluation).

2 Problem Definition

The challenge is that many deformation fields can produce similar-looking registered images, and without constraints, deformations can be anatomically implausible (e.g., tissue folding). VoxelMorph addresses this by amortizing the optimization across a training set, learning a general registration function parameterized by a U-Net CNN.

3 Methodology

3.1 Model Architecture

In this project, based on VoxelMorph, it uses a U-Net-based CNN with an encoder-decoder structure and skip connections. It takes input as a concatenated fixed and moving image pair of shape (2, H, W, D) and outputs a 3-channel velocity field of the same spatial dimensions. We have the encoder that has four downsampling levels with feature channels [16, 32, 32, 32], each using stride-2 convolutions. Then the decoder has a symmetric upsampling with skip connections from the encoder, producing feature channels [32, 32, 32, 32, 16, 16]. The total number of trainable parameters is 250,233.

3.2 Loss Function

The training objective combines two terms:

$$\mathcal{L} = \mathcal{L}_{\text{NCC}} + \lambda \cdot \mathcal{L}_{\text{grad}}, \quad \lambda = 0.5 \tag{1}$$

Normalized Cross-Correlation (NCC): Measures local intensity similarity within a $9 \times 9 \times 9$ sliding window:

$$\mathcal{L}_{\text{NCC}} = -\,\mathbb{E}\left[\frac{(I_f \cdot I_m)^2}{\text{Var}(I_f) \cdot \text{Var}(I_m)}\right] \tag{2}$$

Gradient regularization: Penalizes spatial gradients of the displacement field to encourage smooth deformations:

$$\mathcal{L}_{\text{grad}} = \|\nabla \phi\|^2 \tag{3}$$

3.3 Evaluation Metrics

Dice Similarity Coefficient (DSC): Measures volumetric overlap between the warped moving segmentation and the fixed segmentation:

$$\text{DSC}(A, B) = \frac{2\,|A \cap B|}{|A| + |B|} \tag{4}$$

We evaluate on three FSL brain tissue classes: CSF, Gray Matter, and White Matter. We also report the percentage of voxels with negative Jacobian determinant to measure deformation field regularity.

4 Implementation

4.1 Environment

All experiments used Python 3.10, PyTorch 2.10, and VoxelMorph 0.2 (official PyTorch branch). Training was performed on Google Colab Pro (Tesla T4 GPU, 16 GB VRAM); inference and evaluation on a local Apple Silicon machine.

The original paper used TensorFlow, which is incompatible with Apple Silicon due to a numpy version conflict between TF 2.12 and newer packages. We therefore use the official PyTorch version, which implements an architecturally identical U-Net + STN pipeline. All models were trained from scratch with random initialization; official pretrained weights (.h5 format) are TF-only and were not used.

4.2 Dataset

We use **OASIS-1** [4], consisting of 416 T1-weighted brain MRI subjects aged 18–96. We downloaded disc1 and disc2, yielding **77 preprocessed subjects**.

Preprocessing pipeline:

1. Load Analyze format (.hdr + .img) using nibabel; squeeze extra dims if 4D.
2. Select the MNI-space skull-stripped volume (`*_masked_gfc.img`).
3. Center crop/pad raw shape (176×208×176) to 160×192×224.
4. Min-max intensity normalization to $[0, 1]$.
5. Resize to 128^3 via trilinear interpolation for GPU training.

4.3 Training

Two training runs were conducted, summarized in Table 1.

Table 1: Training Configurations

Version	Subjects	Epochs	GPU Time
v1	39 (disc1)	150	∼1.5 hr
v2	77 (disc1+2)	350	∼6 hr

Optimizer: Adam, learning rate 10^{-4}, batch size 1. Training pairs: $N - 1$ consecutive subject pairs from N volumes. Loss converged from -0.074 (random init) to -0.129 (v2 final epoch).

4.4 Code Structure

- `preprocess.py` — NIfTI loading and intensity normalization
- `load_model.py` — VxmPairwise model initialization
- `train.py` — Training loop with NCC + gradient loss
- `inference_trained.py` — Inference with trained weights
- `evaluate.py` — DSC and Jacobian evaluation pipeline
- `visualize.py` — 6-panel registration visualization

5 Results

5.1 Quantitative Results

Table 2 reports DSC results on 20 test pairs from disc1.

Table 2: DSC Results on 20 Test Pairs

Model	Before	After	Δ	Improved
v1 (39 subj, 150ep)	0.547	0.601	+0.053	20/20
v2 (77 subj, 350ep)	**0.547**	**0.625**	**+0.078**	**20/20**

Per-structure DSC for v2 is shown in Table 3.

Table 3: Per-Structure DSC (v2 Model)

Structure	Before	After	Δ
CSF	0.436	0.489	+0.053
Gray Matter	0.595	0.631	+0.036
White Matter	0.610	0.682	+0.072
Mean	**0.547**	**0.625**	**+0.078**

All 20 test pairs showed improvement under both model versions with no regression cases. White Matter showed the largest improvement (+0.072), consistent with its larger volume and well-defined tissue boundaries.

5.2 Comparison with Original Paper

Table 4: Comparison with Original Paper

Metric	Paper [1]	Ours
Mean DSC (after)	$\sim$0.750	0.625
Training subjects	394	77 (20%)
Label classes	35 (FreeSurfer)	3 (FSL)
Resolution	$160\times192\times224$	128^3
All pairs improved	✓	✓

The DSC gap is expected: we used 20% of the paper's training data, coarser labels (3 vs 35 structures), and lower resolution. The core finding — consistent improvement across all test pairs — is fully reproduced.

5.3 Qualitative Results

Visual inspection confirms the quantitative findings. The registered output closely resembles the fixed image in cortical folding patterns and ventricular shape. The improvement map (difference before minus after registration) is predominantly green across the brain volume, with minor residual misalignment only at ventricular boundaries.

6 Conclusion

We successfully reproduced the VoxelMorph framework for unsupervised deformable brain MRI registration using the official PyTorch implementation trained on OASIS-1 data. Our model achieves a mean DSC of 0.625 after registration compared to 0.547 before, with 20/20 test pairs showing improvement — consistent with the paper's central finding. The gap from the paper's reported 0.750 is explained by reduced training data, coarser evaluation labels, and lower resolution, all of which are documented and quantified. The implementation is fully reproducible from our GitHub repository.

References

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[3] A. V. Dalca, G. Balakrishnan, J. Guttag, and M. R. Sabuncu, "Unsupervised Learning for Fast Probabilistic Diffeomorphic Registration," in *Proc. MICCAI*, 2018, pp. 729–738.

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